

LBP(Labiba And Hebatallah)

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1 The Baseline result

Table 1: Baseline LBP face recognition accuracy under lighting and rotation changes.

Test condition	Accuracy
Normal images	0.975
Uniform brightness change	0.975
Lighting ramp ($s = 0.2$)	0.960
Lighting ramp ($s = 0.4$)	0.945
Lighting ramp ($s = 0.6$)	0.895
Lighting ramp ($s = 0.8$)	0.705
Rotation (5°)	0.970
Rotation (10°)	0.925
Rotation (20°)	0.630

2 Improved Result

3 Improvement Summary

Aspect	Details
Method	Training-gallery augmentation
Added examples	Rotated and relit training photos only
Versions per training photo	25, including the original
Gallery size	200 original; 5,000 improved
Feature extraction	Same LBP descriptor
Matching	Same chi-square distance
Normal accuracy	0.975 before and after
Strong lighting accuracy	$0.705 \rightarrow 0.970$
20° rotation accuracy	$0.630 \rightarrow 0.980$
Tradeoff	Larger gallery; more matching computations

Table 2: LBP recognition accuracy before and after gallery augmentation. The original gallery contains 200 entries; the improved gallery contains 5,000 entries. Gains are reported in percentage points (pp).

Condition	Baseline	Improved	Gain (pp)
Normal photos	0.975	0.975	0.0
Uniform brightness	0.975	0.975	0.0
Lighting ramp ($s = 0.2$)	0.960	0.975	1.5
Lighting ramp ($s = 0.4$)	0.945	0.975	3.0
Lighting ramp ($s = 0.6$)	0.895	0.970	7.5
Lighting ramp ($s = 0.8$)	0.705	0.970	26.5
Rotation (5°)	0.970	0.985	1.5
Rotation (10°)	0.925	0.990	6.5
Rotation (20°)	0.630	0.980	35.0